PARVATHI NALLA

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Education

University of Florida **Aug 2025 – Expected May 2027**
Master of Science in Computer Science *Gainesville, FL*
Relevant Coursework: Advanced Data Structures, Analysis of Algorithms, Distributed Operating Systems, Information Retrieval / NLP

Rajiv Gandhi University of Knowledge Technologies **Aug 2018 – May 2022**
Bachelor of Technology in Computer Science and Engineering *RK Valley, India*

Technical Skills

Programming: Java, Python, C++, Go, JavaScript/TypeScript, SQL, Bash
Web, Backend & Data: React, Angular, Node.js/Express, Django, FastAPI, REST APIs, gRPC, PostgreSQL, MySQL, MongoDB, DynamoDB
Platforms & Quality: Kubernetes, Docker, AWS (Lambda, S3, CloudWatch), Apache Kafka, Git, CI/CD, Jenkins, Unit/Integration Testing, Prometheus, Grafana, ELK
AI/ML: PyTorch, LangChain, OpenAI APIs, FAISS, Retrieval-Augmented Generation (RAG)

Professional Experience

Google | Software Development Intern **May 2026 – Aug 2026**
Sunnyvale, California
- Migrated GKE Gateway Controller components to strongly typed Kubernetes APIs, supporting **10,000+ concurrent routing resources** while removing reflection-heavy dynamic clients.
- Built a translation layer for a caching upgrade affecting **50+ reconcilers**, enabling feature-flagged dual-path testing, rollback, and migration **without production downtime**.
- Refactored Go allocations and eliminated redundant JSON processing, yielding an estimated **40% lower peak heap memory** and **30% faster CPU resynchronization**.

Freightify | Software Development Engineer **Apr 2024 – Jul 2025**
India
- Developed a real-time Apache Kafka and AWS Lambda pipeline, **reducing latency by 25%** while supporting **2× daily data volume**.
- Delivered backend APIs for a freight quotation product used by **5,000+ customers** to compare carrier prices; implemented configurable discount and eligibility rules.
- Standardized unit, integration, contract, and end-to-end testing across **20+ services**, raising **automated test coverage to 85%**.
- Instrumented services with Prometheus, Grafana, SLIs/SLOs, and alerts, **reducing mean time to detect by 47%**.

PromptCloud | Software Development Engineer **Jun 2022 – Apr 2024**
India
- Designed parallel ETL workflows to **increase data throughput by 50%**; optimized processing algorithms to **reduce execution time by 80%**.
- Implemented ELK monitoring across production services, **reducing incident detection time by 65%**; automated deployment tasks to **reduce manual release effort by 30%**.

PromptCloud | Software Development Intern **Nov 2021 – Jun 2022**
India
- Built data-processing pipelines for **750+ GB** of raw data using schema validation, deduplication, and outlier checks.
- Developed crawlers for **50+** JavaScript-rendered websites with rate limiting, proxy rotation, session handling, and failure recovery.

Projects

PointLab (Oct 2026 – Present) *Python, PyTorch, JavaScript | GitHub*
- Developed an interactive 3D LiDAR explorer with geometric patch inspection, masking controls, and reproducible FPS-versus-random sampling benchmarks; includes three research notebooks.

Distributed Social Platform *Gleam, Erlang/OTP, ETS | GitHub*
- Built an actor-based social application with messaging, feeds, voting, OTP supervision for fault tolerance, and benchmarks for up to **1,000 simulated users**.

Research Experience

University of Florida | Graduate Research and Teaching Assistant **Jul 2025 – May 2026**
Gainesville, FL | Financial NLP | Prof. Johannes (Joost) Impink
- Built a Python RAG pipeline using LangChain, OpenAI embeddings, and FAISS to analyze goodwill disclosures from **500+ SEC filings**.
- **Independent extension – Goodwill Atlas:** Developed a FastAPI research application with source-linked extraction, optional FAISS retrieval, and automated evaluation on **36 synthetic labels**. GitHub

University of Florida | Graduate Research and Teaching Assistant **Jul 2025 – May 2026**
Gainesville, FL | Autonomous Driving | Prof. Ivan Ruchkin
- Developed PyTorch/Open3D perception experiments combining RGB images with **32K+ LiDAR points per frame** on nuScenes.